Heterogeneous Green Premiums and Flat Penalties under Credit Constraints and Macroeconomic Shocks: Evidence from London

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1 Introduction

The UK economy experienced severe shocks and shifting paradigms between 2017 and 2024. Following the Russia-Ukraine conflict that began in 2022, energy prices surged (House of Commons 2026), inflation hit double digits and was sustained well above 4% before entering 2024. Bank of England increased Bank Rate from 0.25% in January 2022 to 5.25% in early 2024 (Bank of England 2024), and this posed threats to housing markets by ending the ultra-cheap credit and fundamentally altering market dynamics.

While existing literature documented direct links between macroeconomic variables and aggregate housing markets, there remain gaps in understanding how microeconomic factors behave under such macroeconomic shocks. This study employs a Post-LASSO framework and London property data (2017 and 2024) to analyse how the price associations of micro-level variables, especially energy performance and property characteristics, shift under credit constraints and interest rate spikes. The estimates are conditional associations rather than causal effects; Section 6 discusses the identifying assumptions.

2 Literature Review

Rooted in Rosen' s hedonic pricing theory, property valuation depends on a vector consisting of utility-bearing characteristics (Rosen 1974). An empirical study found that energy performance has a significant positive association with house prices in England and that its marginal effects are non-linear (Fuerst et al. 2015). That study was conducted in a period of low interest rates and low energy costs, so its findings may not hold after the 2022 crisis. Gupta et al. (2022) discovered a "Race-for-Space" in post-COVID era which drastically shifts demand for urban flats to houses in suburban areas, primarily due to existence of "work-from-home" or hybrid working styles and population density in urban areas.

It is theoretically established that the Payment-to-Income (PTI) constraint is the amplifier of impact of interest rate on buyers' purchasing power (Greenwald 2016). Ortalo-Magné

and Rady (2006) proposed the Housing Ladder Model and emphasised that flat buyers are more likely to be first-time buyers (FTB) under credit-constraints who are more vulnerable in interest rate shocks. Ortalo-Magné and Rady (2006) also argued that price levels are driven by actions of marginal buyers choosing from different steps of the ladder, and marginal buyers are forced to step down ladder when upper-ladder properties breach limits. Additionally, increasing real mortgage payments have significant negative impacts on cashflow of credit-constrained households in the UK, simultaneously, households with positive home equity or unused borrowing capacity are less sensitive to such shocks (Campbell and Cocco 2007).

Despite extensive literature on individual factors, most empirical frameworks rely on low-dimensional OLS with manually chosen controls, which may suffer from omitted variable bias when decoding complex feature bundles. By adopting the Post-LASSO framework (Belloni et al. 2014), this essay replaces manual selection with data-driven control selection. This approach is used to estimate how green premiums and flat valuations differ across Housing Affordability Ratio (HAR) regimes (Office for National Statistics 2026) in London between 2017 and 2024.

3 Data and Methodology

This study harmonises Greater London data from the 2017 baseline and 2024 shock to the 2021 LSOA level. The dependent variable is the log-difference of median property prices (HM Land Registry 2026). The feature space incorporates 16 energy factors (Ministry of Housing, Communities and Local Government 2026), seven IMD domains (Ministry of Housing, Communities and Local Government 2025), and PTAL metrics (TfL 2026), including baseline levels, shifts, and intra-LSOA gaps.

After cleaning, 4,358 LSOAs (87.26% of London) are retained in the log-difference sample, which requires complete data and at least five recorded sales in both years. ASHE 2024 does not publish pay growth for the City of London, Harrow or Richmond upon Thames; Inner London growth is used for the City, while Harrow and Richmond are excluded from the 2024 and log-difference samples. Because EPC records are averaged within each LSOA, `is_flat` and `is_abc` are shares of certificates for flats and for bands A–C; their coefficients compare an LSOA in which all such dwellings are flats (or A–C) with one in which none are. Following Belloni et al. (2014), controls are selected separately within the High-HAR and Low-HAR subsamples: LASSO regressions of the outcome, `is_flat` and `is_abc` on the candidate controls are run in each subsample, with penalties chosen by 5-fold cross-validation, and the union of all selected controls is retained. Finally, log-difference (2017–2024) and cross-sectional Post-LASSO OLS models are estimated to describe how the associations of credit constraints and energy performance with prices evolve. The cross-sectional models use the same 15 variables selected by the full-sample log-difference Post-LASSO model, so the control set is held fixed across the two periods. Each cross-sectional model uses all LSOAs with complete data in that year (4,803 in 2017; 4,536 in 2024), whereas the log-difference model requires complete data in both years (4,358).

Note that the dummy variable `is_hhar` is constructed from the 2017 baseline for both 2017 and 2024. It examines how macroeconomic shocks propagate through a pre-existing credit

landscape. Holding the classification fixed avoids mechanical endogeneity from house prices and local HAR moving together, although it does not remove other sources of confounding.

4 Empirical Results

4.1 Cross-Sectional Model: Static Valuation Preferences

As shown in cross-sectional results, the coefficient of `is_abc` increases from a marginally significant 0.085 to a highly significant 0.306. This is consistent with a structural repricing of energy performance, which the market may have undervalued in the credit-easing, low-inflation era. However, the interaction term becomes insignificant in 2024. Additionally, the share of recently built dwellings carries an insignificant 0.067 in 2017 (built from 2007 onwards) and a significant 0.082 in 2024 (built from 2012 onwards); because the two cohorts are defined differently, these coefficients are not directly comparable. These transitions are consistent with energy cost shocks repricing the energy-efficient feature bundle.

Table 1: Cross-Sectional OLS Regression Models: Common Feature Set

	(1) 2017 Model	(2) 2024 Model
Intercept	11.600***	12.340***
	(0.116)	(0.173)
`is_abc`	0.085*	0.306***
	(0.050)	(0.053)
`is_abc:is_hhar`	0.130**	-0.105
	(0.066)	(0.073)
`is_flat`	-0.108**	-0.159***
	(0.045)	(0.043)
`is_flat:is_hhar`	0.258***	0.227***
	(0.056)	(0.069)
`is_hhar`	0.226***	0.146***
	(0.027)	(0.036)
Selected Controls		
Crime Index	-0.009	-0.065***
Employment Index	1.007***	0.396
Environment Index	0.003***	0.005***
Income (£)	0.000***	0.000***
Number of Rooms	0.092***	0.152***
Observations	4,803	4,536
R^2	0.717	0.629

Note: $^{*}p < 0.1$;$^{**}p < 0.05$;$^{***}p < 0.01$. Clustered SE by LAD in parentheses.

Term `is_flat` shows a stronger discount in Low-HAR areas, but the flat-premium in High-HAR areas is smaller in 2024 than 2017. In 2024, Low-HAR flats are discounted by approximately 14.7% ($e^{-0.159} - 1$), but the positive interaction term results in a 7.0% premium ($e^{-0.159+0.227} - 1$) over local houses within High-HAR areas. Additionally, note the positive coefficient of High-HAR houses reduces from 0.226 to 0.146.

4.2 Log-Difference Model: Cumulative Growth Heterogeneity

Table 2: Post-LASSO Log Difference Model (Full Sample)

Variable	Log Difference Model
Intercept	0.080
	(0.122)
`is_abc`	0.100***
	(0.036)
`is_abc:is_hhar`	-0.122***
	(0.041)
`is_flat`	-0.071***
	(0.027)
`is_flat:is_hhar`	-0.015
	(0.029)
`is_hhar`	-0.092***
	(0.019)
Control Variables	Included (20+ features)
Observations	4,358
R^2	0.127
Adjusted R^2	0.123
F Statistic	41.887***

Note: $^{*}p < 0.1;^{**}p < 0.05;^{***}p < 0.01$. Clustered SE by LAD in parentheses.

In the log-difference model, `is_abc` is associated with approximately 10.5% higher price growth in Low-HAR areas ($e^{0.100}-1$), but with approximately 2.2% lower growth ($e^{0.100-0.122}-1$) in High-HAR areas, relative to local Non-ABC properties.

In contrast to energy efficiency, being a flat is, counterintuitively, associated with lower price growth. In the log-difference model regression, `is_flat` has a statistically significant coefficient of -0.071. This suggests that, on average the median flat price growth in the same Low-HAR LSOA is lower than houses by approximately 6.9% ($e^{-0.071}-1$) between 2017 and 2024, ceteris paribus. In addition, cumulative price growth of houses in High-HAR areas is significantly smaller than ones in Low-HARs by 8.8% ($e^{-0.092}-1$).

Finally, High-HAR flats' growth is lower than Low-HAR houses' growth by 16.3% ($e^{-0.071-0.015-0.092}-1$), but the High-HAR penalty of 1.5% is not statistically significant.

The R^2 of 0.127 should not be too worrying as cross-sectional price-level variations are absorbed, while the F-statistic confirms joint significance.

5 Mechanism Discussion

5.1 Main Source of Divergence

The divergence in the flat coefficient across HAR regimes likely arises from the asymmetric transmission of credit frictions through PTI. The mechanisms below are interpretations consistent with the estimates, not tested causal channels. Payment-to-Income ratio can be decomposed

into:

$$PTI = HAR \times LTV \times c(r)$$

HAR amplifies the partial effect of interest rate on PTI. If HAR is higher, then unit-increase in interest rate has a higher impact on PTI than the Low-HAR case.

5.2 PTI and Green Premium

One major distinction across HAR areas is budgetary slack. Given an income, Low-HAR households, facing lower repayment pressures, are more likely to have financial capacity to convert future energy savings into current purchasing power. Conversely, consistent with Ortalo-Magné and Rady (2006), marginal buyers in High-HAR areas are usually more credit constrained.

Although the 2024 cross-sectional interaction term is not statistically significant, the disappearance of the 2017 premium suggests that high-HAR buyers may be hitting PTI limits. This interpretation is supported by the significant growth deficit in the log-difference model. As rates increase, even if high-HAR buyers rationally recognise the future savings of energy-efficient properties, they may be financially excluded from purchasing them. This distinction would create an asymmetric pattern in prices, as Low-HAR buyers may push demand (and the green premium) up while High-HAR buyers' demand diminishes.

5.3 PTI, Housing Ladder and Flat Penalty

Similar setup may explain the flat market. Flat prices in Low-HAR areas are 14.7% ($e^{-0.159} - 1$) lower than Low-HAR houses, but the gap reverses in High-HAR areas to 7.0% after controlling for other variables.

In Low-HAR areas, the budgetary slack allows households to have a wider range of choice by having a higher mortgage potential. By the housing ladder model and "race-for-space" hypothesis, households may tend to purchase houses, if available, to enjoy larger areas and outdoor spaces. Thus, in Low-HAR areas, demand may be driven primarily by demand for space, and flats may need to sell at a discount to attract buyers.

In structurally high-priced areas, house prices may push first-time buyers (FTBs) beyond lenders' PTI affordability limits, excluding them from houses. For these marginal buyers, when upper-ladder properties are unreachable, they move down the ladder to purchase flats to secure a foothold in their preferred locations, creating an inelastic demand. This structural "down-laddering" may create a persistent price floor for flats, which would explain the intra-regional flat premiums observed in both the 2017 baseline (+16.2%, $e^{-0.108+0.258} - 1$) and the 2024 shock period (+7.0%, $e^{-0.159+0.227} - 1$).

While the price floor holds in the cross-section, down-laddering does not appear to protect these High-HAR properties over time. In the log-difference model, the −8.8% coefficient on `is_hhar` indicates the vulnerability of High-HAR areas to the interest rate shock. High-HAR flats perform worst (−16.3%), consistent with the flat penalty compounding with the sensitivity of High-HAR credit constraints.

Crucially, while this price floor cushions flats in the cross-section, the premium was compressed over the period that includes the 2022–2024 rate shock. From 2017 to 2024, the High-HAR flat premium shrank from 16.2% to 7.0%, suggesting that the interest rate rise eroded rather than inflated localised price premia. The log-difference model points in the same direction: the baseline flat growth penalty (-6.9%, $e^{-0.071}-1$), the insignificant interaction (-0.015) and the spatial baseline penalty (-8.8%, $e^{-0.092}-1$) combine to a 16.3% ($e^{-0.071-0.015-0.092}-1$) growth deficit for High-HAR flats. This pattern suggests that tighter credit conditions disproportionately eroded entry-level, highly leveraged assets, weakening the support previously provided by down-laddering.

Furthermore, the temporal dynamics of our results are illuminated by the evolving structure of the UK mortgage market. Unlike the adjustable-rate dominance described by Campbell and Cocco (2007), the majority of contemporary residential loans are fixed-rate (80–95% per FCA (2026)). This structural shift may introduce a lagged refinancing shock; buyers who locked in low rates between 2017 and 2020 could face a sharp escalation in servicing costs upon expiry after 2021. In High-HAR districts, this liquidity squeeze could be particularly severe, potentially triggering forced divestment waves, exerting downward pressure on property valuations. This would be consistent with the growth deficits we observe for energy-efficient properties and flats in High-HAR areas.

6 Limitations

Five limitations qualify the results. First, the estimates are conditional associations. Double selection protects inference on the selected coefficients from model-selection error, but it does not address confounding: the design relies on unconfoundedness given the selected controls, and there is no alternative identification strategy (such as an instrument or a difference-in-differences design) to fall back on if that assumption fails. Second, controls are selected separately within each HAR group, but they enter the final OLS additively, so their coefficients are restricted to be equal across groups; the interactions with `is_hhar` are also not treated as separate selection targets. Penalties are chosen by cross-validation rather than the plug-in rule of Belloni et al. (2014). Third, heterogeneity is studied along a single, pre-chosen dimension (the High/Low-HAR split). Effects may vary along other dimensions that this design cannot detect; data-driven methods such as causal forests would be needed to search for such heterogeneity. Fourth, median prices use all Price Paid records, including non-standard (Category B) transactions, so a few LSOAs have implausible medians (for example, below £10,000 or above £40 million in 2017). The five-sale threshold limits their influence in the log-difference model, but the cross-sectional samples have no such restriction, and the exclusion of Harrow and Richmond may make the sample less representative of higher-income outer London. Fifth, in the selection step the shares `is_flat` and `is_abc` are converted to integers, so the treatment equations select controls that predict whether an LSOA consists entirely of flats (or A–C dwellings) rather than the shares themselves. The results have not been re-estimated with these issues corrected.

7 Conclusion

This study documents heterogeneity in the price associations of flats and energy efficiency across credit-constraint regimes in London, during a dual shock of rising energy costs and interest rates. Using HAR as a proxy for credit constraints, the results are consistent with PTI acting as a filter in property pricing: credit-constrained buyers appear to be excluded from energy-efficient properties, flats hold a cross-sectional price floor, yet flats show weaker price growth over time. Overall, the evidence suggests that London's pricing is shifting from property quality towards affordability. If so, PTI constraints make outcomes divergent and unequal, allowing households with budgetary slack to capture energy savings and space, while credit-constrained households are left holding less efficient, depreciating assets.

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A Decomposition of Payment-to-Income Ratio

To analyse how credit constraints vary across regions, decompose the Payment-to-Income (PTI) ratio.

First, define the following variables: P as the median property price, I as the annual income, L as the loan amount, M as the annual mortgage payment, and r as the interest rate.

HAR is Housing Affordability Ratio, and LTV is Loan-to-Value Ratio.

The key ratios are defined as $PTI = M/I$, $HAR = P/I$, and $LTV = L/P$.

By definition:

$$PTI = \frac{M}{I} \tag{1}$$

Since the annual payment M is the product of the loan L and the mortgage constant $c(r)$:

$$PTI = \frac{L \times c(r)}{I} \tag{2}$$

Given that the total loan L equals $P \times LTV$:

$$PTI = \frac{P \times LTV \times c(r)}{I} \tag{3}$$

Rearrange the terms:

$$PTI = \left(\frac{P}{I}\right) \times LTV \times c(r) \tag{4}$$

By substituting $HAR = P/I$, we get to:

$$PTI = HAR \times LTV \times c(r) \tag{5}$$